Chapter 4

Exponents and Logarithms

Very large or very small numbers or expressions can easily be expressed using exponents. As a result, calculations and solution of mathematical problems become easier. Scientific or standard form of a number is expressed using exponents. Therefore, every student should have the knowledge of exponents and its applications.

Multiplication and division of numbers or expressions and exponent related calculations have become easier with the help of logarithms. Use of logarithm in scientific calculations was the only way before calculators and computers became available. Still the use of logarithm is important as the alternative of calculator and computer.

In this chapter, exponents and logarithms have been discussed in detail.

At the end of the chapter, the students will be able to —

- Explain the rational exponent.
- explain and apply the positive integral exponents, zero and negative integral exponents
- solve problems by describing and applying the rules of exponents
- explain the n th root and rational fractional exponents and express the n th root in terms of exponents
- explain logarithms
- prove and apply the formulae of logarithms
- explain natural logarithm and common logarithm
- explain the scientific form of numbers
- explain the characteristic and mantissa of common logarithm
- calculate common and natural logarithm by calculator

Exponents or Indices

In class VI, we have got the idea of exponents and in class VII, we have learnt the exponential rules for multiplication and division. Expressions associated with exponent and base are called **exponential expression**.

Work: Fill in the blanks

Successive multiplication of the same expression	Exponential expression	base	power or exponent
$2 \times 2 \times 2$	2^3	2	3
$3 \times 3 \times 3 \times 3$		3	
$a \times a \times a$	a^3		
$b \times b \times b \times b \times b$			5

If a is any real number, successive multiplication of n as is a^n; that is, $a \times a \times a \times \ldots \times a$ (n times a) $= a^n$, where n is a positive integer. Here, n is **index or power** and a is **base**. Conversely $a^n = a \times a \times a \times \ldots \times a$ (n times a).

Exponents may not only be positive integers, they may be negative integers or positive fractions or negative fractions. That is, for base $a \in R$ (set of real numbers) and power $n \in Q$ (set of rational numbers), a^n is defined. The case of $n \in N$ (set of natural numbers) is especially considered. Besides, exponents may also be irrational. But as it is out of curriculum, it has not been discussed in this chapter.

Index Laws

Let, $a \in R$ (set of real numbers) and $m, n \in N$ (set of natural numbers).

Formula 1 (Multiplication). $a^m \times a^n = a^{m+n}$

Formula 2 (Division). $\dfrac{a^m}{a^n} = \begin{cases} a^{m-n} \text{ when } m \geq n \\ \dfrac{1}{a^{n-m}} \text{ when } n > m \end{cases}$

Fill in the blanks of the following table:

$a \neq 0, m > n \qquad m = 5, n = 3$	$a \neq 0, n > m \qquad m = 3, n = 5$
$a^5 \times a^3 = (a\times a\times a\times a\times a)\times(a\times a\times a)$ $= a\times a\times a\times a\times a\times a\times a\times a = a^8 = a^{5+3}$	$a^3 \times a^5 =$
$\frac{a^5}{a^3} =$	$\frac{a^3}{a^5} = \frac{a\times a\times a}{a\times a\times a\times a\times a} = \frac{1}{a^2} = \frac{1}{a^{5-3}}$

$\therefore$ In general $a^m \times a^n = a^{m+n}$ and $\frac{a^m}{a^n} = \begin{cases} a^{m-n} \text{ when } m \geq n \\ \frac{1}{a^{n-m}} \text{ when } n > m \end{cases}$

Formula 3 (power of product). $(ab)^n = a^n \times b^n$

We observe, $(5 \times 2)^3 = (5 \times 2) \times (5 \times 2) \times (5 \times 2)$ [$\because a^3 = a \times a \times a,\ a = 5 \times 2$]
$= (5 \times 5 \times 5) \times (2 \times 2 \times 2)$
$= 5^3 \times 2^3$

In general, $(ab)^n = ab \times ab \times ab \times \ldots \times ab$ [Successive multiplication of n ab's]
$= (a \times a \times a \times \ldots \times a) \times (b \times b \times b \times \ldots \times b)$
$= a^n \times b^n$

Formula 4 (power of quotient). $\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n},\ (b \neq 0)$

We observe, $\left(\frac{5}{2}\right)^3 = \frac{5}{2} \times \frac{5}{2} \times \frac{5}{2} = \frac{5^3}{2^3}$

In general, $\left(\frac{a}{b}\right)^n = \frac{a}{b} \times \frac{a}{b} \times \frac{a}{b} \ldots \times \frac{a}{b}$ [Successive multiplication of n times $\frac{a}{b}$]
$= \frac{a \times a \times a \times \ldots \times a}{b \times b \times b \times \ldots \times b} = \frac{a^n}{b^n}$

Formula 5 (power of power). $(a^m)^n = a^{mn}$

$(a^m)^n = a^m \times a^m \times a^m \times \ldots \times a^m$ [Successive multiplication of n times a^m]
$= a^{m+m+m\ldots+m}$ [in the power, multiplication of n times of exponent]
$= a^{m\times n} = a^{mn}$

$\therefore (a^m)^n = a^{mn}$

Zero and Negative Indices

For 0 and negative indices a^0 and a^{-n} are defined as follows (where n is natural number).

Definition 1 (zero index). $a^0 = 1,\ (a \neq 0)$

Definition 2 (negative index). $a^{-n} = \frac{1}{a^n},\ (a \neq 0, n \in N)$

For all integer indices m and n the definition $\frac{a^m}{a^n} = a^{m-n}$ holds true. Observe, $\frac{a^n}{a^n} = a^{n-n} = a^0$

But $\frac{a^n}{a^n} = \frac{a \times a \times a \times \ldots \times a \quad (n \text{ times})}{a \times a \times a \times \ldots \times a \quad (n \text{ times})} = 1$

$\therefore a^0 = 1$

and $\frac{1}{a^n} = \frac{a^0}{a^n} = a^{0-n} = a^{-n}$

Example 1. Find the values: 1) $\frac{5^2}{5^3}$ 2) $\left(\frac{2}{3}\right)^5 \times \left(\frac{2}{3}\right)^{-5}$

Solution:

1) $\frac{5^2}{5^3} = 5^{2-3} = 5^{-1} = \frac{1}{5^1} = \frac{1}{5}$

2) $\left(\frac{2}{3}\right)^5 \times \left(\frac{2}{3}\right)^{-5} = \left(\frac{2}{3}\right)^{5-5} = \left(\frac{2}{3}\right)^0 = 1$

Example 2. Simplify: 1) $\frac{5^4 \times 8 \times 16}{2^5 \times 125}$ 2) $\frac{3 \cdot 2^n - 4 \cdot 2^{n-2}}{2^n - 2^{n-1}}$

Solution:

1) $\frac{5^4 \times 8 \times 16}{2^5 \times 125} = \frac{5^4 \times 2^3 \times 2^4}{2^5 \times 5^3} = \frac{5^4 \times 2^{3+4}}{5^3 \times 2^5} = \frac{5^4}{5^3} \times \frac{2^7}{2^5}$

$= 5^{4-3} \times 2^{7-5} = 5^1 \times 2^2 = 5 \times 4 = 20$

2) $\frac{3 \cdot 2^n - 4 \cdot 2^{n-2}}{2^n - 2^{n-1}} = \frac{3 \cdot 2^n - 2^2 \cdot 2^{n-2}}{2^n - 2^n \cdot 2^{-1}} = \frac{3 \cdot 2^n - 2^{2+n-2}}{2^n - 2^n \cdot \frac{1}{2}}$

$= \frac{3 \cdot 2^n - 2^n}{\left(1 - \frac{1}{2}\right) \cdot 2^n} = \frac{(3-1) \cdot 2^n}{\frac{1}{2} \cdot 2^n} = \frac{2 \cdot 2^n}{\frac{1}{2} \cdot 2^n} = 2 \cdot 2 = 4$

Example 3. Prove that, $(a^p)^{q-r} \cdot (a^q)^{r-p} \cdot (a^r)^{p-q} = 1$

Solution: $(a^p)^{q-r} \cdot (a^q)^{r-p} \cdot (a^r)^{p-q} = a^{p(q-r)} \cdot a^{q(r-p)} \cdot a^{r(p-q)}$ $[\because (a^m)^n = a^{mn}]$

$= a^{pq-pr} \cdot a^{qr-pq} \cdot a^{pr-qr} = a^{pq-pr+qr-pq+pr-qr} = a^0 = 1$

Work: Fill in the blanks:
1) $3 \times 3 \times 3 \times 3 = 3^{\square}$ 2) $5^{\square} \times 5^3 = 5^5$ 3) $a^2 \times a^{\square} = a^{-3}$
4) $(-5)^0 = \square$ 5) $\frac{4}{4^{\square}} = 1$

n th Root of a

We observe, $5^{\frac{1}{2}} \times 5^{\frac{1}{2}} = \left(5^{\frac{1}{2}}\right)^2$

Again, $5^{\frac{1}{2}} \times 5^{\frac{1}{2}} = 5^{\frac{1}{2}+\frac{1}{2}} = 5^{2\times\frac{1}{2}} = 5$

$\therefore \left(5^{\frac{1}{2}}\right)^2 = 5$

Square of $(5^{\frac{1}{2}}) = 5$ and square root (second root) of $5 = 5^{\frac{1}{2}}$

$5^{\frac{1}{2}}$ is written as $\sqrt{5}$.

Again we observe, $5^{\frac{1}{3}} \times 5^{\frac{1}{3}} \times 5^{\frac{1}{3}} = \left(5^{\frac{1}{3}}\right)^3$

Again, $5^{\frac{1}{3}} \times 5^{\frac{1}{3}} \times 5^{\frac{1}{3}} = 5^{\frac{1}{3}+\frac{1}{3}+\frac{1}{3}} = 5^{3\times\frac{1}{3}} = 5$

Cube (power 3) of $5^{\frac{1}{3}} = 5$ and cube root (third root) of $5 = 5^{\frac{1}{3}}$

$5^{\frac{1}{3}}$ is written as $\sqrt[3]{5}$.

In the case of n th root,

$a^{\frac{1}{n}} \times a^{\frac{1}{n}} \times a^{\frac{1}{n}} \times \ldots \times a^{\frac{1}{n}}$ [successive multiplication of n $a^{\frac{1}{n}}$'s] $= \left(a^{\frac{1}{n}}\right)^n$

Again, $a^{\frac{1}{n}} \times a^{\frac{1}{n}} \times a^{\frac{1}{n}} \times \ldots \times a^{\frac{1}{n}}$

$= a^{\frac{1}{n}+\frac{1}{n}+\frac{1}{n}+\ldots+\frac{1}{n}}$ [In the exponent, sum of n $\frac{1}{n}$'s]

$= a^{n\times\frac{1}{n}} = a$

$\therefore \left(a^{\frac{1}{n}}\right)^n = a$

The n th power of $a^{\frac{1}{n}}$ is a and the n th root of a is $a^{\frac{1}{n}}$

i.e, n th power of $a^{\frac{1}{n}}$ is $\left(a^{\frac{1}{n}}\right)^n = a$ and n th root of a is $(a)^{\frac{1}{n}} = a^{\frac{1}{n}} = \sqrt[n]{a}$

The n th root of a is written as $\sqrt[n]{a}$.

Example 4. Simplify: 1) $(12)^{-\frac{1}{2}} \times \sqrt[3]{54}$ 2) $(-3)^3 \times \left(-\frac{1}{2}\right)^2$

Solution:

1) $(12)^{-\frac{1}{2}} \times \sqrt[3]{54} = \dfrac{1}{(12)^{\frac{1}{2}}} \times (54)^{\frac{1}{3}} = \dfrac{1}{(2^2 \times 3)^{\frac{1}{2}}} \times (3^3 \times 2)^{\frac{1}{3}}$

$= \dfrac{1}{(2^2)^{\frac{1}{2}} \times 3^{\frac{1}{2}}} \times (3^3)^{\frac{1}{3}} \cdot 2^{\frac{1}{3}} = \dfrac{1}{2 \cdot 3^{\frac{1}{2}}} \times (3^3 \times 2)^{\frac{1}{3}}$

$= \dfrac{2^{\frac{1}{3}}}{2^1} \times \dfrac{3^1}{3^{\frac{1}{2}}} = \dfrac{3^{1-\frac{1}{2}}}{2^{1-\frac{1}{3}}} = \dfrac{3^{\frac{1}{2}}}{2^{\frac{2}{3}}} = \dfrac{3^{\frac{1}{2}}}{4^{\frac{1}{3}}} = \dfrac{\sqrt{3}}{\sqrt[3]{4}}$

2) $(-3)^3 \times \left(-\dfrac{1}{2}\right)^2 = (-3)(-3)(-3) \times \left(-\dfrac{1}{2}\right)\left(-\dfrac{1}{2}\right) = -27 \times \dfrac{1}{4} = -\dfrac{27}{4}$

Work: Simplify: 1) $\dfrac{2^4 \cdot 2^2}{32}$ 2) $\left(\dfrac{2}{3}\right)^{\frac{5}{2}} \times \left(\dfrac{2}{3}\right)^{-\frac{5}{2}}$ 3) $8^{\frac{3}{4}} \div 8^{\frac{1}{2}}$

To be noticed:

1) Under the condition $a > 0,\ a \neq 1$ if $a^x = a^y$ then $x = y$

2) Under the condtion $a > 0,\ b > 0,\ x \neq 0$ if $a^x = b^x$ then $a = b$

Example 5. Solve: $4^{x+1} = 32$

Solution: $4^{x+1} = 32$ or, $(2^2)^{x+1} = 32$ or, $2^{2x+2} = 2^5$

$\therefore 2x + 2 = 5$ [if $a^x = a^y$, $x = y$]

or, $2x = 5 - 2$ or, $2x = 3$

$\therefore x = \dfrac{3}{2}$

Exercises 4.1

Simplify (1 - 8):

1. $\dfrac{7^3 \times 7^{-3}}{3 \times 3^{-4}}$
2. $\dfrac{\sqrt[3]{7^2} \cdot \sqrt[3]{7}}{\sqrt{7}}$
3. $(2^{-1} + 5^{-1})^{-1}$
4. $(2a^{-1} + 3b^{-1})^{-1}$
5. $\left(\dfrac{a^2 b^{-1}}{a^{-2} b}\right)^2$
6. $\sqrt{x^{-1}y} \cdot \sqrt{y^{-1}z} \cdot \sqrt{z^{-1}x}$ $(x > 0,\ y > 0,\ z > 0)$
7. $\dfrac{2^{n+4} - 4 \cdot 2^{n+1}}{2^{n+2} \div 2}$
8. $\dfrac{3^{m+1}}{(3^m)^{m-1}} \div \dfrac{9^{m+1}}{(3^{m-1})^{m+1}}$

Prove (9 - 15):

9. $\dfrac{4^n-1}{2^n-1}=2^n+1$

10. $\dfrac{2^{2p+1}\cdot 3^{2p+q}\cdot 5^{p+q}\cdot 6^p}{3^{p-2}\cdot 6^{2p+2}\cdot 10^p\cdot 15^q}=\dfrac{1}{2}$

11. $\left(\dfrac{a^l}{a^m}\right)^n\cdot\left(\dfrac{a^m}{a^n}\right)^l\cdot\left(\dfrac{a^n}{a^l}\right)^m=1$

12. $\dfrac{a^{p+q}}{a^{2r}}\times\dfrac{a^{q+r}}{a^{2p}}\times\dfrac{a^{r+p}}{a^{2q}}=1$

13. $\left(\dfrac{x^a}{x^b}\right)^{\frac{1}{ab}}\cdot\left(\dfrac{x^b}{x^c}\right)^{\frac{1}{bc}}\cdot\left(\dfrac{x^c}{x^a}\right)^{\frac{1}{ca}}=1$

14. $\left(\dfrac{x^a}{x^b}\right)^{a+b}\cdot\left(\dfrac{x^b}{x^c}\right)^{b+c}\cdot\left(\dfrac{x^c}{x^a}\right)^{c+a}=1$

15. $\left(\dfrac{x^p}{x^q}\right)^{p+q-r}\cdot\left(\dfrac{x^q}{x^r}\right)^{q+r-p}\cdot\left(\dfrac{x^r}{x^p}\right)^{r+p-q}=1$

16. If $a^x=b$, $b^y=c$ and $c^z=a$, then show that, $xyz=1$

Solve(17 - 20):

17. $4^x=8$

18. $2^{2x+1}=128$

19. $(\sqrt{3})^{x+1}=(\sqrt[3]{3})^{2x-1}$

20. $2^x+2^{1-x}=3$

21. $P=x^a$, $Q=x^b$ and $R=x^c$

    1) Find the values of $P^{bc}\cdot Q^{-ca}$.

    2) Find the values of $\left(\dfrac{P}{Q}\right)^{a+b}\times\left(\dfrac{Q}{R}\right)^{b+c}\div 2(RP)^{a-c}$

    3) Show that, $\left(\dfrac{P}{Q}\right)^{a^2+ab+b^2}\times\left(\dfrac{Q}{R}\right)^{b^2+bc+c^2}\times\left(\dfrac{R}{P}\right)^{c^2+ca+a^2}=1$

22. $X=(2a^{-1}+3b^{-1})^{-1}$, $Y=\sqrt[pq]{\dfrac{x^p}{x^q}}\times\sqrt[qr]{\dfrac{x^q}{x^r}}\times\sqrt[rp]{\dfrac{x^r}{x^p}}$

    and $Z=\dfrac{5^{m+1}}{(5^m)^{m-1}}\div\dfrac{25^{m+1}}{(5^{m-1})^{m+1}}$, where $x,p,q,r>0$

    1) Find the value of X

    2) Show that, $Y+\sqrt[3]{81}=5$

    3) Show that, $Y\div Z=25$

Logarithms

Logarithms are used to find the values of exponential expressions. Logarithm is written in brief as **Log**. Product, quotient, etc. of large numbers or quantities can easily be determined with the help of log.

We know, $2^3 = 8$; this mathematical statement is written in terms of log as $\log_2 8 = 3$. Again, conversely, if $\log_2 8 = 3$, it can be written in terms of exponents as $2^3 = 8$. That is, if $2^3 = 8$ then $\log_2 8 = 3$ and conversely, if $\log_2 8 = 3$ then $2^3 = 8$. Similarly, $2^{-3} = \frac{1}{2^3} = \frac{1}{8}$ can be written in terms of log as, $\log_2 \frac{1}{8} = -3$.

If $a^x = N, (a > 0,\ a \neq 1)$ then $x = \log_a N$ is defined as a **based** $log_a N$

Note: If $a > 0$ then a^x is always positive whatever may be the values of x, positive or negative. So, only the log of positive numbers has values that are real; log of zero or negative numbers have no real value.

Work: Express the following values in exponents in terms of log:

In terms of exponent	in terms of log
$10^2 = 100$	
$3^{-2} = \frac{1}{9}$	
$2^{\frac{1}{2}} = \sqrt{2}$	
$2^{-\frac{1}{2}} = \frac{1}{\sqrt{2}}$	
$\sqrt[4]{2^4} = 2$	

In terms of exponent	in terms of log
$10^0 = 1$	$\log_{10} 1 = 0$
$e^0 = \ldots$ $a^0 = 1$	$\log_e 1 = \ldots$ $\ldots = \ldots$
$10^1 = 10$	$\log_{10} 10 = 1$
$e^1 = \ldots$	$\ldots = \ldots$
$\ldots = \ldots$	$\log_a a = 1$

Laws of Logarithms

Let, $a > 0,\ a \neq 1;\ b > 0,\ b \neq 1$ and $M > 0,\ N > 0$

Formula 6 (zero and one log). If $a > 0,\ a \neq 1$ 1) $\log_a 1 = 0$ 2) $\log_a a = 1$

Proof: We know from the formula of exponents, $a^0 = 1$

$\therefore$ from the definition of log, we get, $\log_a 1 = 0$ (proved)

Again, we know, from the formula of exponents, $a^1 = a$

$\therefore$ from the definition of log, we get, $\log_a a = 1$ (proved)

Formula 7 (Log of products). $\log_a(MN) = \log_a M + \log_a N$

Proof: Let, $\log_a M = x,\ \log_a N = y$

$\therefore M = a^x,\ N = a^y$

Now, $MN = a^x \cdot a^y = a^{x+y}$

$\therefore \log_a(MN) = x + y$

or, $\log_a(MN) = \log_a M + \log_a N$ [putting the values of x, y]

$\therefore \log_a(MN) = \log_a M + \log_a N$ (proved)

Note: $\log_a(MNP\ldots) = \log_a M + \log_a N + \log_a P + \ldots$

Note: $\log_a(M \pm N) \neq \log_a M \pm \log_a N$

Formula 8 (log of quotient). $\log_a \frac{M}{N} = \log_a M - \log_a N$

Proof: Let, $\log_a M = x,\ \log_a N = y$

$\therefore M = a^x,\ N = a^y$

Now, $\frac{M}{N} = \frac{a^x}{a^y} = a^{x-y}$

$\therefore \log_a \frac{M}{N} = x - y$

$\therefore \log_a \frac{M}{N} = \log_a M - \log_a N$ (proved)

Formula 9 (Log of power). $\log_a M^r = r\log_a M$

Proof: Let, $\log_a M = x,\ \therefore M = a^x$

or, $(M)^r = (a^x)^r$ or, $M^r = a^{rx}$

$\therefore \log_a M^r = rx$ or, $\log_a M^r = r\ \log_a M$

$\therefore \log_a M^r = r\log_a M$ (proved)

Note: $(\log_a M)^r$ and $r\log_a M$ may not be equal.

For example, $(\log_2 4)^5 = (\log_2 2^2)^5 = 2^5 = 32,\ 5\log_2 4 = 5 \cdot 2 = 10 \neq 32$

Formula 10 (change of base). $\log_a M = \log_b M \times \log_a b$

Proof: Let, $\log_a M = x,\ \log_b M = y$

$\therefore a^x = M,\ b^y = M$

$\therefore a^x = b^y$ or, $(a^x)^{\frac{1}{y}} = (b^y)^{\frac{1}{y}}$ or, $b = a^{\frac{x}{y}}$

$\therefore \frac{x}{y} = \log_a b$ or, $x = y\log_a b$

or, $\log_a M = \log_b M \times \log_a b$ (proved)

Corollary 1. $\log_a b = \frac{1}{\log_b a}$ or $\log_b a = \frac{1}{\log_a b}$

Proof: We know, $\log_a M = \log_b M \times \log_a b$

Putting $M = a$ we get, $\log_a a = \log_b a \times \log_a b$

or, $1 = \log_b a \times \log_a b$

$\therefore \log_a b = \frac{1}{\log_b a}$ or $\log_b a = \frac{1}{\log_a b}$ (proved)

Example 6. Find the values: 1) $\log_{10}100$ 2) $\log_3\frac{1}{9}$ 3) $\log_{\sqrt{3}}81$

Solution:

1) $\log_{10}100 = \log_{10}10^2 = 2\log_{10}10$ $[\because \log_{10}M^r = r\log_{10}M]$
   $= 2 \times 1 = 2$ $[\because \log_a a = 1]$
2) $\log_3\left(\frac{1}{9}\right) = \log_3\left(\frac{1}{3^2}\right) = \log_3 3^{-2} = -2\log_3 3$ $[\because \log_a M^r = r\log_a M]$
   $= -2 \times 1 = -2$ $[\because \log_a a = 1]$
3) $\log_{\sqrt{3}}81 = \log_{\sqrt{3}}3^4 = \log_{\sqrt{3}}\{(\sqrt{3})^2\}^4 = \log_{\sqrt{3}}(\sqrt{3})^8$
   $= 8\log_{\sqrt{3}}\sqrt{3} = 8 \times 1 = 8$ $[\because \log_a a = 1]$

Example 7. 1) What is the log of $5\sqrt{5}$ to the base of 5? 2) If the log of 400 is 4 then what is the base of log?

Solution:

1) Log of $5\sqrt{5}$ to the base of 5
   $= \log_5 5\sqrt{5} = \log_5(5 \times 5^{\frac{1}{2}}) = \log_5 5^{\frac{3}{2}}$
   $= \frac{3}{2}\log_5 5$ $[\because \log_a M^r = r\log_a M]$
   $= \frac{3}{2} \times 1 = \frac{3}{2}$ $[\because \log_a a = 1]$
2) Let the base be a
   $\therefore$ By the question, $\log_a 400 = 4$
   $\therefore a^4 = 400$ or, $a^4 = (20)^2 = \{(2\sqrt{5})^2\}^2 = (2\sqrt{5})^4$

$\therefore a = 2\sqrt{5} \qquad [\because a^x = b^x,\ a^x \neq 0,\ a = b]$

$\therefore$ Base $2\sqrt{5}$

Example 8. Find the value of x: 1) $\log_{10} x = -2$ 2) $\log_x 324 = 4$

Solution:

1) $\log_{10} x = -2$

or, $x = 10^{-2} = \dfrac{1}{10^2} = \dfrac{1}{100} = 0.01$

$\therefore x = 0.01$

2) $\log_x 324 = 4$

or, $x^4 = 324 = 3 \times 3 \times 3 \times 3 \times 2 \times 2 = 3^4 \times 2^2$

or, $x^4 = 3^4 \times (\sqrt{2})^4$

or, $x^4 = (3\sqrt{2})^4$

$\therefore x = 3\sqrt{2}$

Example 9. Prove that, $3\log_{10} 2 + \log_{10} 5 = \log_{10} 40$

Solution: Left hand side $= 3\log_{10} 2 + \log_{10} 5$

$= \log_{10} 2^3 + \log_{10} 5 \qquad [\because \log_a M^r = r\log_a M]$

$= \log_{10} 8 + \log_{10} 5$

$= \log_{10}(8 \times 5) \qquad [\because \log_a (MN) = \log_a M + \log_a N]$

$= \log_{10} 40 =$ right hand side (proved)

Example 10. Simplify: $\dfrac{\log_{10}\sqrt{27} + \log_{10} 8 - \log_{10}\sqrt{1000}}{\log_{10} 1.2}$

Solution: $\dfrac{\log_{10}\sqrt{27} + \log_{10} 8 - \log_{10}\sqrt{1000}}{\log_{10} 1.2}$

$= \dfrac{\log_{10}(3^3)^{\frac{1}{2}} + \log_{10} 8 - \log_{10}(10^3)^{\frac{1}{2}}}{\log_{10}\dfrac{12}{10}}$

$= \dfrac{\log_{10} 3^{\frac{3}{2}} + \log_{10} 2^3 - \log_{10}(10)^{\frac{3}{2}}}{\log_{10} 12 - \log_{10} 10}$

$$= \frac{\frac{3}{2}\log_{10}3 + 3\log_{10}2 - \frac{3}{2}\log_{10}10}{\log_{10}(3 \times 2^2) - \log_{10}10}$$

$$= \frac{\frac{3}{2}(\log_{10}3 + 2\log_{10}2 - 1)}{\log_{10}3 + 2log_{10}2 - 1} \qquad [\because \log_{10}10 = 1]$$

$$= \frac{3}{2}$$

Exercises 4.2

1. Find the values:
   1) $\log_3 81$ 2) $\log_5 \sqrt[3]{5}$ 3) $\log_4 2$
   4) $\log_{2\sqrt{5}} 400$ 5) $\log_5(\sqrt[3]{5} \cdot \sqrt{5})$
2. Find the value of x :
   1) $\log_5 x = 3$ 2) $\log_x 25 = 2$ 3) $\log_x \frac{1}{16} = -2$
3. Show that,
   1) $5\log_{10}5 - \log_{10}25 = \log_{10}125$
   2) $\log_{10}\frac{50}{147} = \log_{10}2 + 2\log_{10}5 - \log_{10}3 - 2\log_{10}7$
   3) $3\log_{10}2 + 2\log_{10}3 + \log_{10}5 = \log_{10}360$
4. Simplify:
   1) $7\log_{10}\frac{10}{9} - 2\log_{10}\frac{25}{24} + 3\log_{10}\frac{81}{80}$
   2) $\log_7(\sqrt[5]{7} \cdot \sqrt{7}) - \log_3\sqrt[3]{3} + \log_4 2$
   3) $\log_e\frac{a^3b^3}{c^3} + \log_e\frac{b^3c^3}{d^3} + \log_e\frac{c^3d^3}{a^3} - 3\log_e b^2c$
5. $x = 2, y = 3, z = 5, w = 7$
   1) What is the log of $\sqrt{y^3}$ to the base 3.
   2) Find the value of $w\log\frac{xz}{y^2} - x\log\frac{z^2}{x^2y} + y\log\frac{y^4}{x^4z}$
   3) Show that, $\frac{\log\sqrt{y^3} + y\log x - \frac{y}{x}\log(xz)}{\log(xy) - \log z} = \log_y\sqrt{y^3}$

Scientific or Standard Form of Numbers

We can express very large numbers or very small numbers easily using exponents.

Such as, the velocity of light = 300000 km/sec = 300000000 m/sec

$= 3 \times 100000000$ m/sec $= 3 \times 10^8$ m/sec

Again, the radius of a hydrogen atom

$= 0.0000000037$ cm

$= \dfrac{37}{10000000000}$ cm $= 37 \times 10^{-10}$ cm

$= 3.7 \times 10 \times 10^{-10}$ cm $= 3.7 \times 10^{-9}$ cm

For convenience, very large or very small numbers are expressed in the form $a \times 10^n$ where, $1 \leq a < 10$ and $n \in Z$ The number of the form $a \times 10^n$ is called the **scientific or standard form** of number.

Work: Express the following numbers in scientific form :

1) 15000 2) 0.000512 3) 123.000512

Logarithmic Methods

Logarithmic systems are of two kinds :

1) **Natural Logarithm:** Mathematician John Napier (1550-1617) of Scotland published the first book on logarithms in 1614 by taking e as its base. e is an irrational number, $e = 2.71828\ldots$. Such logarithms are called Napierian logarithms **Napierian logarithm** or e **based logarithm** or **natural logarithm**. $\log_e x$ is also written in the form $\ln x$.

2) **Common Logarithm:** Mathematician Henry Briggs (1561-1630) of England prepared a **table of logarithm** (**log table**) in 1624 with 10 as the base. This logarithm is called **Briggs logarithm** or 10 **based logarithm** or **practical logarithm**. This logarithm is written as $\log_{10} x$.

Note: If there is no mention of base, in case of algebraic expressions e and in case of numbers 10, are considered the base. In log table 10 is taken as the base.

Characteristics of Common Log

Let a number N be expressed in the scientific form as

$N = a \times 10^n$, where $N > 0, 1 \leq a < 10$ and $n \in Z$

Taking base 10 log of both sides,

$\log_{10} N = \log_{10}(a \times 10^n) = \log_{10} a + \log_{10} 10^n = \log_{10} a + n\log_{10} 10$

$\therefore \log_{10} N = n + \log_{10} a$ [$\because \log_{10} 10 = 1$]

Suppressing the base 10, we have, $\log N = n + \log a$

n is called the characteristic of $\log N$.

Note: It is evident from the table that characteristic of a number having m digits to the left of decimal point is 1 less than the number of digits to the left of decimal sign.

N	Scientific form of N	Exponent	Number of digits on the left of the decimal point	Characteristic
6237	6.237×10^3	3	4	$4 - 1 = 3$
623.7	6.237×10^2	2	3	$3 - 1 = 2$
62.37	6.237×10^1	1	2	$2 - 1 = 1$
6.237	6.237×10^0	0	1	$1 - 0 = 0$
0.6237	6.237×10^{-1}	-1	0	$0 - 1 = -1$

Note: Now observe from the following table: If there is no integral part of a number and the number has k 0's immediately after decimal point, then the characteristic of the logarithm of the number is $\{-(k+1)\}$.

If any characteristic is negative, '$-$' sign is not placed on the left of the characteristic, rather '$-$' (bar sign) over the characteristic. Such as, characteristic -3 will be written as $\bar{3}$. Otherwise, whole part of the log including mantissa will be negative.

N	Scientific form of N	Exponent	Number of zeroes between decimal point and its next first significant digit	Characteristic
0.6237	6.237×10^{-1}	-1	0	$-(0+1) = -1 = \bar{1}$
0.06237	6.237×10^{-2}	-2	1	$-(1+1) = -2 = \bar{2}$
0.006237	6.237×10^{-3}	-3	2	$-(2+1) = -3 = \bar{3}$

Note: Characteristic may be either positive or negative, but mantissa will always be positive.

Example 11. Find the characteristics of log of the following numbers :

1) 5570 2) 45.70 3) 0.4305 4) 0.000435

Solution:

1) $5570 = 5.570 \times 1000 = 5.570 \times 10^3$

$\therefore$ Characteristic of log of the number is 3

The number of digits in the number 5570 is 4.

$\therefore$ Characteristic of log of the number is $= 4 - 1 = 3$

2) $45.70 = 4.570 \times 10^1$

$\therefore$ Characteristic of log of the number is 1

There are 2 digits in the integral part (i.e. on the left of the decimal point) of the number.

$\therefore$ Characteristic of log of the number is $= 2 - 1 = 1$

3) $0.4305 = 4.305 \times 10^{-1}$ $\therefore$ Characteristic of log of the number is -1

$\therefore$ Characteristic of log of the number is $= 0 - 1 = -1 = \bar{1}$

There is no zero in between decimal point and its next first significant digit of the number 0.4305, i.e. there is 0 zeroes.

$\therefore$ Characteristic of log of the number is $= -(0+1) = -1 = \bar{1}$

$\therefore$ Characteristic of log of the number 0.4305 is $\bar{1}$

4) $0.000435 = 4.35 \times 10^{-4}$

$\therefore$ Characteristic of log of the number is -4 or $\bar{4}$

$\therefore$ Characteristic of log of the number is $= -(3+1) = -4 = \bar{4}$

$\therefore$ Characteristic of log of the number 0.000435 is $\bar{4}$

Mantissa of Common Log

Mantissa of Common Logarithm of any number is a non-negative number less than 1. It is mainly an irrational number. But the value of mantissa is determined upto a certain decimal place. Mantissa of the log of a number can be found from the log table. Again, it can also be found by calculator. We shall find the mantissa of the log of any number using 2nd method, that is by calculator.

Example 12. Find the characteristic and the mantissa of log2717 :

Solution: Let us use the calculator: $\boxed{AC}\ \boxed{\log}\ \boxed{2717}\ \boxed{=}$ 3.43409

$\therefore$ Characteristic of log2717 is 3 and the mantissa is .43409

Example 13. Find the characteristic and the mantissa of log43.517.

Solution: Let us use the calculator: $\boxed{AC}\ \boxed{\log}\ \boxed{43.517}\ \boxed{=}$ 1.63866

$\therefore$ Characteristic of log43.517 is 1 and mantissa is .63866

Example 14. Find the characteristic and mantissa of 0.00836?

Solution: Let us use the calculator: $\boxed{AC}\ \boxed{\log}\ \boxed{0.00836}\ \boxed{=}$ -2.07779

$-2.07779 = -3 + 0.92221 = \bar{3}.92221$

$\therefore$ Characteristic of log0.00836 is -3 and the mantissa is .92221.

Example 15. Find $\log_e 10$:

Solution: $\log_e 10 = \dfrac{1}{\log_{10} e} = \dfrac{1}{\log_{10} 2.71828} = \dfrac{1}{0.43429}$ [using the calculator]

$= 2.30259$ (approx)

Alternative : We use the calculator : $\boxed{AC}\ \boxed{\ln}\ \boxed{10}\ \boxed{=}$ 2.30259

Work: Compute base 10 and base e logarithm of the following numbers using calculator:

1) 2550 2) 52.143 3) 0.4145 4) 0.0742

Exercises 4.3

1. On what condition $a^0 = 1$?
   1) $a = 0$ 2) $a \neq 0$ 3) $a > 0$ 4) $a \neq 1$
2. Which one of the following is the value of $\sqrt[3]{5} \cdot \sqrt[3]{5}$?
   1) $\sqrt[6]{5}$ 2) $(\sqrt[3]{5})^3$ 3) $(\sqrt{5})^3$ 4) $\sqrt[3]{25}$
3. On what exact condition $\log_a a = 1$?
   1) $a > 0$ 2) $a \neq 1$ 3) $a > 0, a \neq 1$ 4) $a \neq 0, a > 1$
4. If $\log_x 4 = 2$, what is the value of x?
   1) 2 2) ± 2 3) 4 4) 10
5. Under what condition can a number be written in the form $a \times 10^n$?
   1) $1 < a < 10$ 2) $1 \leq a \leq 10$ 3) $1 \leq a < 10$ 4) $1 < a \leq 10$
6. If $a > 0,\ b > 0$ and $a \neq 1,\ b \neq 1$
   (i) $\log_a b \times \log_b a = 1$
   (ii) $\log_a M^r = M \log_a r$
   (iii) $\log_a(\sqrt[3]{a} \cdot \sqrt{a}) = \frac{5}{6}$
   Which of the above information are correct?
   1) i 2) ii 3) i and iii 4) ii and iii
7. What is the characteristic of the common log of 0.0035?
   1) 3 2) 1 3) $\bar{2}$ 4) $\bar{3}$

Considering the number 0.0225 answer the following questions (8 – 10):

8. Which one of the following is in the form a^n?

1) $(2.5)^2$ 2) $(.015)^2$ 3) $(1.5)^2$ 4) $(.15)^2$

9. Which one of the following is the scientific form of the number ?
   1) 225×10^{-4} 2) 22.5×10^{-3} 3) 2.25×10^{-2} 4) $.225 \times 10^{-1}$

10. What is the characteristic of the common log of the number ?
    1) $\bar{2}$ 2) $\bar{1}$ 3) 0 4) 2

11. Express into scientific form:
    1) 6530 2) 60.831 3) 0.000245 4) 37500000
    5) 0.00000014

12. Express in ordinary decimals:
    1) 10^5 2) 10^{-5} 3) 2.53×10^4 4) 9.813×10^{-3}
    5) 3.12×10^{-5}

13. Find the characteristics of common logarithm of the following numbers (without using calculator) :
    1) 4820 2) 72.245 3) 1.734 4) 0.045
    5) 0.000036

14. Find the characteristics and mantissa of the common logarithm of the following numbers using calculator:
    1) 27 2) 63.147 3) 1.405 4) 0.0456
    5) 0.000673

15. Find the common logarithm of the product/quotient (approximate value upto five decimal places) :
    1) 5.34×8.7 2) 0.79×0.56 3) $22.2642 \div 3.42$
    4) $0.19926 \div 32.4$

16. If $\log 2 = 0.30103$, $\log 3 = 0.47712$ and $\log 7 = 0.85410$, find the value of the following expressions :
    1) $\log 9$ 2) $\log 28$ 3) $\log 42$

17. Given, $x = 1000$ and $y = 0.0625$
    1) Express x in the form $a^n b^n$ where, a and b are prime numbers.
    2) Express the product of x and y in scientific form.
    3) Find the characteristic and mantissa of the common logarithm of xy